

Helena Vela Beltran (Do IT Now)

EESSI hackathon - 25 September 2026 @ EuroHPC User Days

<https://eessi.io/docs/training-events/2026/hackathon-eurohpc-user-days>

Prepared environment for hands-on

- Small Rocky 10 cluster (in the cloud)
- **You need to create an account!**
 - Signup: <https://mokey.tutorial.eessi.science/auth/signup>
 - Accounts will only be approved for access on the day of the tutorial, so **please record your username/password !**
 - “Reset password” link does **not** work
- Access via ssh or web browser (**pick one and stick to it**)
 - Shell access: `ssh tutorial.eessi.science`
 - Use login node for hands-on, it has 16 cores so should be fine to share for small group
 - Via browser: <https://tutorial.eessi.science/>
 - Make sure to change default “Time” to 4 hours
 - Take 8 cores
- System will be up for the day



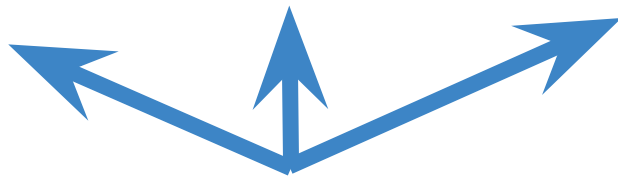
hackmd.io/@multixscale/EESSI

This could be you! #1 - Software Developers, Operations Staff



- You might start developments on your computer
- Complexity grows, with it the need of a proper environment
 - HPC Cluster
 - Test systems
 - CI/CD pipeline
- Or the moment to test in production comes

This could be you! #2 - European Researchers



1 project, multiple allocations
across HPC Machines

The landscape of scientific computing has changed...

- **Explosion of available scientific software applications** (bioinformatics, AI boom, ...)
- Increasing interest in **cloud** for scientific computing (flexibility!)
- Increasing **variety in processor (micro)architectures** beyond Intel & AMD:
Arm is ~~coming~~ here (see Fugaku, JUPITER, ...), RISC-V is coming (soon?)
- In stark contrast: available (wo)manpower in HPC support teams is (still) limited
- **Software installation tools like EasyBuild and Spack are no longer sufficient...**



*What if you no longer have to install
a **broad range of scientific software**
from scratch on every laptop, HPC cluster,
or cloud instance you use or maintain,
without compromising on performance?*



Q&A via EESSI Slack

Please join the #eurohpc-user-days-2026-hackathon channel in the EESSI Slack for questions and discussion

Step 1) **Join the EESSI Slack**, see “Slack channel” link at <https://eessi.io>

Step 2) **Join [#eurohpc-user-days-2026-hackathon](https://eurohpc-user-days-2026-hackathon) channel** in EESSI Slack
(direct link: <https://eessi-hpc.slack.com/archives/C0C3TJ0RCG7>)



Agenda



- [09:00] - Welcome
- [09:30 -10:00] - Quick intro to EESSI
- [10:00 - 10:30] - **Hands-on:** playing with EESSI as a User
- [10:30 - 11:00] - Coffee Break
- [11:00 - 12:00] - Building on top of EESSI and how to contribute it
- [12:00 - 13:00] - Lunch Break (Group picture!)

Agenda



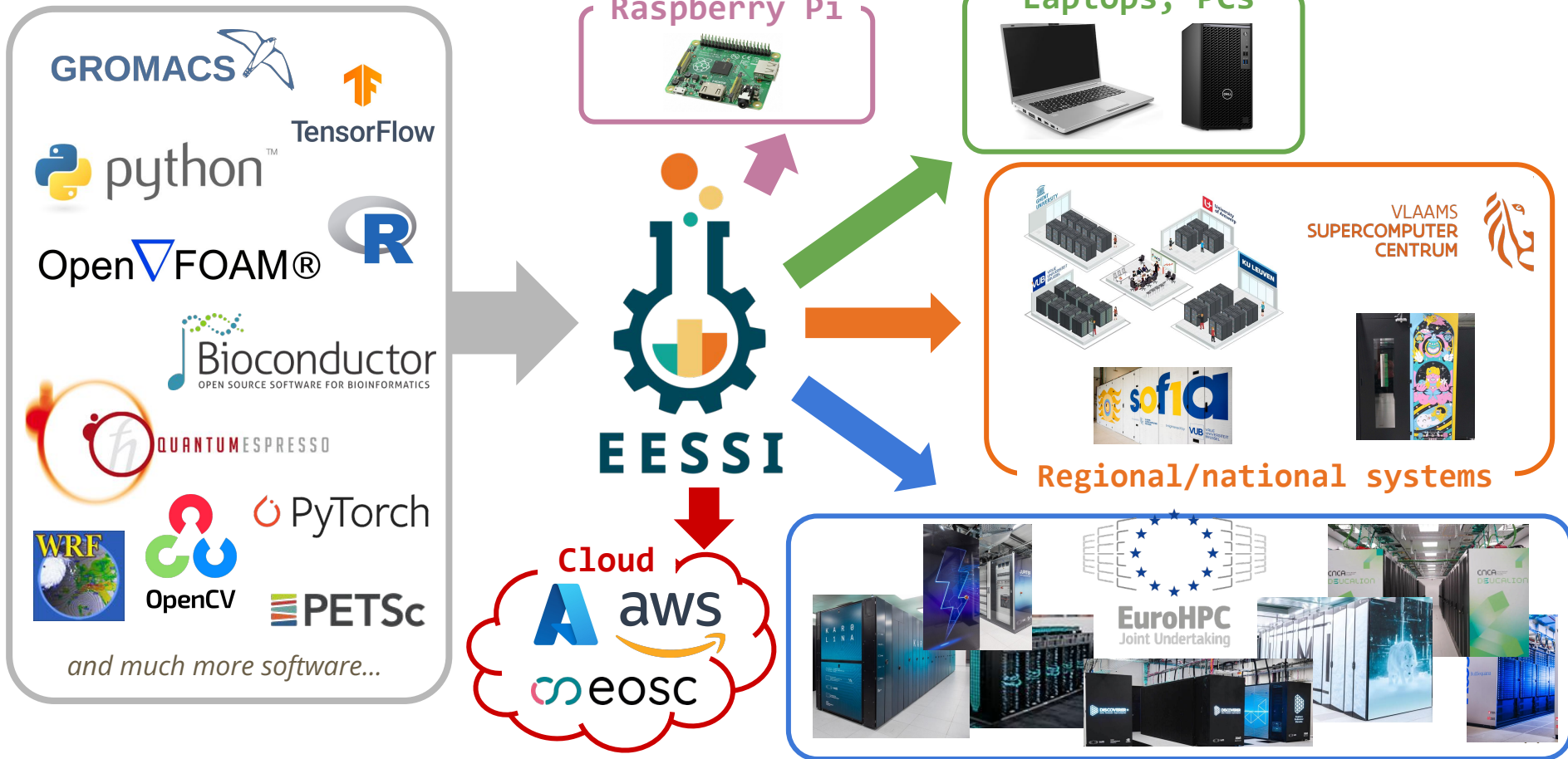
- [13:00-15:00] **Hands-on:** exercises, onboarding software into EESSI
- [15:00-15:30] Coffee Break
- [15:30-17:00] **Hands-on** (continued)
- [17:00-17:30] Show & Tell

Major goals of EESSI



- Providing a truly **uniform software stack**
 - Use the (exact) same software environment everywhere
 - **Without sacrificing performance** for “mobility of compute” (like is typically done with containers/conda)
- **Avoid duplicate work** (for researchers, HPC support teams, sysadmins, ...)
 - Tools that automate software installation process (EasyBuild, Spack) are not sufficient anymore
 - Go beyond sharing build recipes => work towards a shared software stack
- Facilitate HPC training, development of (scientific) software, ...

EESSI as shared software stack for HPC, cloud, and beyond



Accessing EESSI via CernVM-FS



```
# Native installation
# Installation commands for RHEL-based distros
# like CentOS, Rocky Linux, AlmaLinux, Fedora, ...

# install CernVM-FS

sudo yum install -y
https://ecsft.cern.ch/dist/cvmfs/cvmfs-release/cvmfs-release-latest.noarch.rpm
sudo yum install -y cvmfs

# create client configuration file for CernVM-FS
# (no proxy, 10GB local CernVM-FS client cache)
sudo bash -c "echo 'CVMFS_CLIENT_PROFILE='single'' > /etc/cvmfs/default.local"
sudo bash -c "echo 'CVMFS_QUOTA_LIMIT=10000' >> /etc/cvmfs/default.local"

# Make sure that EESSI CernVM-FS repository is accessible
sudo cvmfs_config setup
```

Alternative ways of accessing EESSI are available, via a container image, via cvmfsexec, ...
eessi.io/docs/getting_access/native_installation - eessi.io/docs/getting_access/eessi_container

Using EESSI

eessi.io/docs/using_eessi/eessi_demos



```
/cvmfs/software.eessi.io/versions/2025.06/software
```

```
`-- linux
  |-- aarch64
  |   |-- generic
  |   |-- neoverse_n1
  |   |-- neoverse_v1
  |   |-- a64fx
  |   |-- nvidia
  |       |-- grace
  |-- x86_64
  |   |-- amd
  |   |   |-- zen2
  |   |   |-- zen3
  |   |   |-- zen4
  |   |   |-- zen5
  |   |-- generic
  |-- intel
  |   |-- haswell
  |   |-- cascadelake
  |   |-- icelake
  |   |-- sapphirerapids
  |   |-- skylake_avx512
  |       |-- modules
  |       |-- software
```

```
$ source /cvmfs/software.eessi.io/versions/2025.06/init/lmod/bash
Modules purged before initialising EESSI
Module for EESSI/2025.06 loaded successfully
```

EESSI has selected x86_64/amd/zen2 as the compatible CPU target for EESSI/2025.06 **Automatically detects CPU microarchitecture**

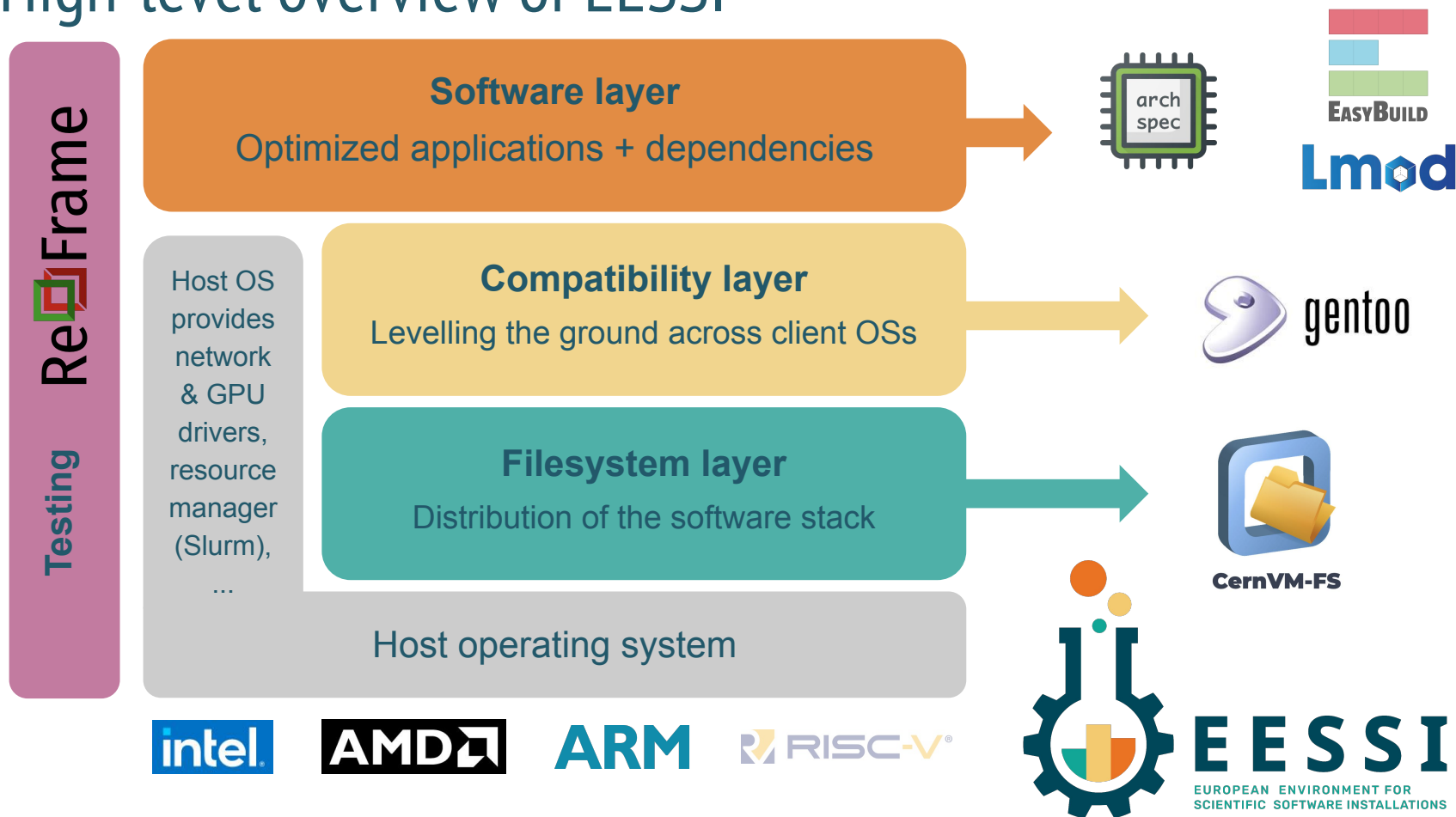
EESSI did not identify an accelerator on the system
(for debug information when loading the EESSI module, set the environment variable EESSI_MODULE_DEBUG_INIT)

```
{EESSI 2025.06} $ module load R/4.5.1-gfbbf-2025a
```

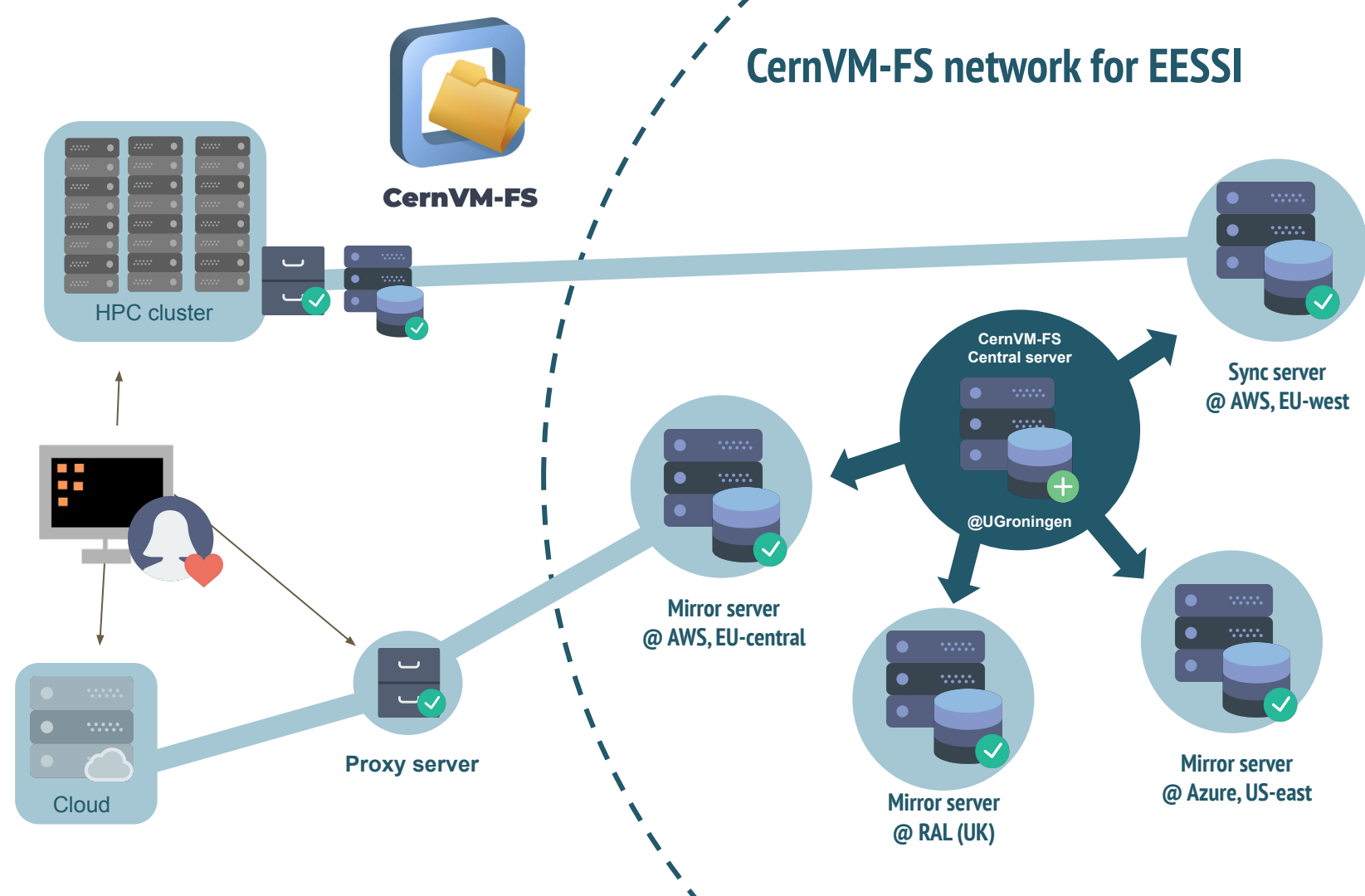
```
{EESSI 2025.06} $ which R
/cvmfs/software.eessi.io/versions/2025.06/software/linux/x86_64/
amd/zen2/software/R/4.5.1-gfbbf-2023a/bin/R
```

```
{EESSI 2025.06} $ R --version
R version 4.5.1
```

High-level overview of EESSI



CernVM-FS network for EESSI



New releases and developments

New releases and repositories

software.eessi.io (Production Repositories)

dev.eessi.io

2023.06

Since June 2023

1,247 installations
(total: ~17,800)

679 unique software
projects

Toolchain:
foss/(2022b+23a+23b)
→ **GCC 12.4** ↓

2025.06

Current default

1,275 installations
(total: ~21,400)

640 unique software
projects

Toolchain:
foss/(2024a+25a+25b)
→ **GCC 13.3** ↑

2026.06

Baby repo! 🚀

250 installations
(total: ~4,500)

240 unique software
projects

Toolchain:
foss/(2026.1)
→ **GCC 15.2** ↑

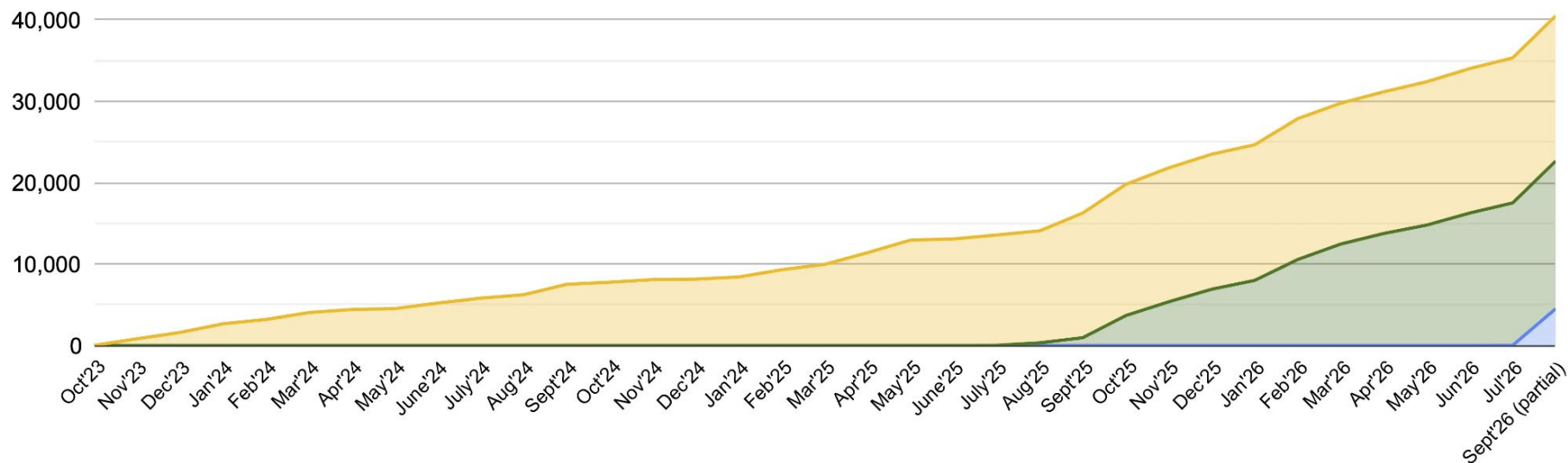
- Developer repository
- Staging area for upcoming releases
- [More about it later :\)](#)

New software evolution - all stacks



Number of software installations per EESSI version in software.eessi.io repository (CPU+GPU)

2023.06 2025.06 2026.06



Overview of available software

An up-to-date overview of available software is available in the EESSI documentation:
https://eessi.io/docs/available_software

Software available in EESSI

Overview of software available in [EESSI's production repository](https://eessi.io) `software.eessi.io`.

752 unique software projects (+ 3007 unique extensions)

name:GROMACS

GROMACS

[\(more details\)](#)

<https://www.gromacs.org>

GROMACS is a versatile package to perform molecular dynamics, i.e. simulate the Newtonian equations of motion for systems with hundreds to millions of particles. This is a CPU only build, containing both MPI and threadMPI binaries for both single and double precision. It also contains the gmxml extension for the single precision MPI build.

Available in EESSI versions: 2023.06 2025.06

Supported CPU families: AMD Intel Arm

Supported GPU families: NVIDIA



GROMACS

GROMACS is a versatile package to perform molecular dynamics, i.e. simulate the Newtonian equations of motion for systems with hundreds to millions of particles.

This is a GPU enabled build, containing both MPI and threadMPI binaries.

It also contains the gmxml extension for the single precision MPI build.

homepage: <https://www.gromacs.org>

Available installations

GROMACS version	Supported CPU targets	Supported GPU targets	EESSI version	Module
2025.4	generic: aarch64, x86_64 Arm : a64fx, neoverse_n1, neoverse_v1, nvidia/grace AMD : zen2, zen3, zen4 Intel : haswell, skylake_avx512, sapphirerapids, icelake, cascadelake	(none)	2025.06	GROMACS/2025.4-foss-2025b
2025.2	generic: aarch64, x86_64 Arm : a64fx, neoverse_n1, neoverse_v1, nvidia/grace AMD : zen2, zen3, zen4 Intel : haswell, skylake_avx512, sapphirerapids, icelake, cascadelake	(none)	2025.06	GROMACS/2025.2-foss-2025a
2024.4	generic: aarch64, x86_64 Arm : neoverse_n1,	NVIDIA : cc70, cc80, cc90	2023.06	GROMACS/2024.4-foss-2023b-

(new) CPU targets

aarch64		generic a64fx neoverse_n1 neoverse_v1
	nvidia	grace
	aws	<i>graviton4</i>
x86_64		generic
	amd	zen2 zen3 zen4 zen5
	intel	haswell skylake_avx512 cascadelake icelake sapphirerapids <i>graniterapids</i>



green -> new CPU targets
in EESSI 2026.06

bold -> from the
beginning of the project

*experimental in EESSI 2026.06

riscv64	generic
---------	---------

Supported GPU targets (NVIDIA)

https://eessi.io/docs/software_layer/gpu_targets



- Different generations of NVIDIA GPUs, based on CUDA Compute Capability (CC), see also <https://developer.nvidia.com/cuda/gpus>
 - In EESSI 2026.06: 16 CPU targets x 5 CUDA compute capabilities = 80 CPU-GPU combos

CPU microarchitecture	CUDA compute capability					
	7.0	7.5	8.0	9.0	10.0	12.0
aarch64/ generic	2023.06.x 2025.06.x	2026.06.x	2023.06.x 2025.06.x 2026.06.x	2023.06.x 2025.06.x 2026.06.x	2025.06.x 2026.06.x	2025.06.x 2026.06.x
aarch64/a64fx	-	-	-	-	-	-
aarch64/aws/ graviton4	-	2026.06.x	2026.06.x	2026.06.x	2026.06.x	2026.06.x
aarch64/ neoverse_n1	2023.06.x 2025.06.x	2026.06.x	2023.06.x 2025.06.x 2026.06.x	2023.06.x 2025.06.x 2026.06.x	2025.06.x 2026.06.x	2025.06.x 2026.06.x

- GPU driver libraries (include `libcuda.so`) must be exposed to EESSI
- See https://www.eessi.io/docs/site_specific_config/gpu

See also *NVIDIA + AMD GPU support in EESSI* talk @ EUM'26 ([slides](#), [recording](#))

Supported GPU targets (AMD/ROCm) https://eessi.io/docs/software_layer/gpu_targets



- First set of modules available in EESSI/2025.06

```
----- /cvmfs/software.eessi.io/versions/2025.06/software/linux/x86_64/generic/accel/amd/gfx908/modules/all -----
Boost/1.88.0-rocm-compilers-19.0.0-ROCm-6.4.1          (g,D)   rocBLAS/4.4.0-rocm-compilers-19.0.0-ROCm-6.4.1          (g)
HIP/6.4.1-rocm-compilers-19.0.0-ROCm-6.4.1             (g)     rocm-cmake/0.14.0-rocm-compilers-19.0.0-ROCm-6.4.1      (g)
hipBLAS-common/1.2.0-rocm-compilers-19.0.0-ROCm-6.4.1 (g)     rocm-compilers/19.0.0-ROCm-6.4.1                        .
hipBLASLt/0.12.1-rocm-compilers-19.0.0-ROCm-6.4.1     (g)     ROCm-LLVM/19.0.0-GCCcore-14.2.0-ROCm-6.4.1             .
HIPIFY/19.0.0-rocm-compilers-19.0.0-ROCm-6.4.1        (g)     rocminfo/1.0.0-rocm-compilers-19.0.0-ROCm-6.4.1         (g)
msgpack-cxx/7.0.0-rocm-compilers-19.0.0-ROCm-6.4.1    (g)     rocTracer/4.1.0-rocm-compilers-19.0.0-ROCm-6.4.1        (g)
RCCL/2.22.3-rocm-compilers-19.0.0-ROCm-6.4.1          (g)     Tensile/4.43.0-rocm-compilers-19.0.0-ROCm-6.4.1         (g)
```

- Support for different AMD GPU generations: [see docs for details](#)

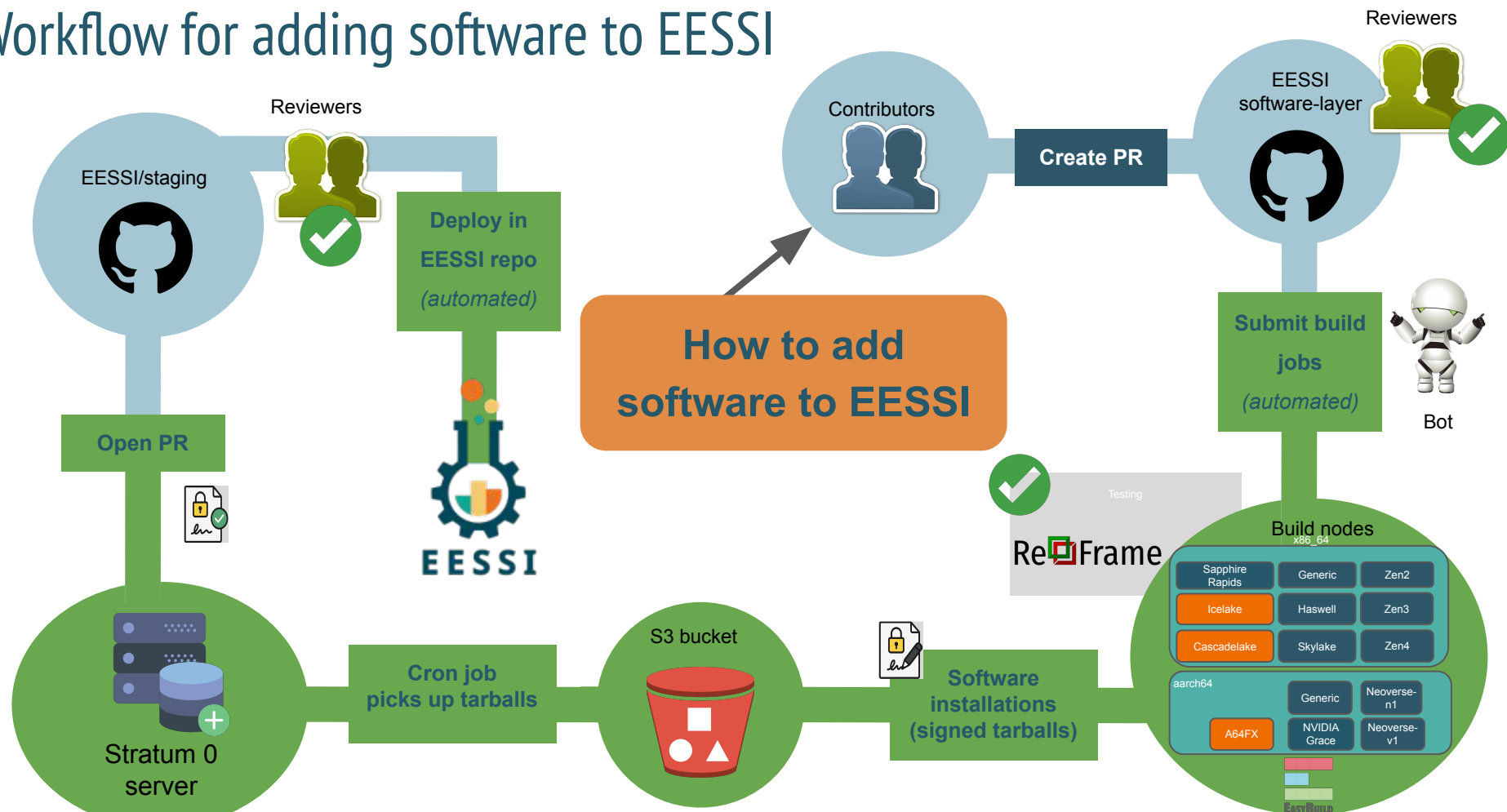
gfx908: MI100, gfx90a: MI200, gfx942: MI300, gfx1030: RX6000, gfx1100/1101:

RX7000, gfx1200/1201: RX8000 and RX9000

- Initial goal: GROMACS with ROCm 6.4.1 (WIP), ROCm 7.x next

See also *NVIDIA + AMD GPU support in EESSI* talk @ EUM'26 ([slides](#), [recording](#))

Workflow for adding software to EESSI



Build-test-deploy bot

- Build/Deploy/Ingest Strategy - Fully automatic or with human in the loop
- Interaction model: Plain CI, external CLI, GitHub App
 - GitHub App / scripts that understand(s) structured commands in PR comments, labels, and merge/close
- Whole CI infrastructure underneath:
 - GitHub runners, self-hosted runners → HPC clusters (on-prem/Cloud), each running its own bot instance
- Bot only defines interface, repo owners define the actual build & test logic



oaisa commented [last week](#)

Memt

```
bot: build repo:eessi.io-2025.06-software instance:eessi-bot-mc-aws for:arch=x86_64/amd/zen3
bot: build repo:eessi.io-2025.06-software instance:eessi-bot-mc-aws for:arch=x86_64/amd/zen4
bot: build repo:eessi.io-2025.06-software instance:eessi-bot-mc-aws for:arch=x86_64/intel/skylake_avx512
bot: build repo:eessi.io-2025.06-software instance:eessi-bot-mc-aws for:arch=x86_64/intel/haswell
bot: build repo:eessi.io-2025.06-software instance:eessi-bot-mc-aws for:arch=x86_64/intel/icelake
bot: build repo:eessi.io-2025.06-software instance:eessi-bot-mc-aws for:arch=x86_64/intel/cascadelake
bot: build repo:eessi.io-2025.06-software instance:eessi-bot-mc-aws for:arch=x86_64/intel/sapphirerapids
bot: build repo:eessi.io-2025.06-software instance:eessi-bot-mc-aws for:arch=x86_64/generic
bot: build repo:eessi.io-2025.06-software instance:eessi-bot-mc-aws for:arch=aarch64/generic
bot: build repo:eessi.io-2025.06-software instance:eessi-bot-mc-aws for:arch=aarch64/neoverse_n1
bot: build repo:eessi.io-2025.06-software instance:eessi-bot-mc-aws for:arch=aarch64/neoverse_v1
```



eessi-bot-aws bot commented [last week](#) • edited ▾

New job on instance eessi-bot-mc-aws for repository eessi.io-2025.06-software
Building on: amd-zen3
Building for: x86_64/amd/zen3
Job dir: /project/def-users/SHARED/jobs/2026.01/pr_1357/122018

date	job status	comment
Jan 15 11:22:04 UTC 2026	submitted	job id 122018 awaits release by job manager
Jan 15 11:23:02 UTC 2026	released	job awaits launch by Slurm scheduler
Jan 15 11:38:24 UTC 2026	running	job 122018 is running
Jan 15 13:32:50 UTC 2026	finished	► 😊 SUCCESS (click triangle for details)
Jan 15 13:32:50 UTC 2026	test result	► 😊 SUCCESS (click triangle for details)

Software testing is an important part of EESSI



- Smoke test: sanity check commands are run by EasyBuild to check that installed software is not horribly broken while using EasyBuild
- Regression testing via EESSI test suite: eessi.io/docs/test-suite
 - Collection of portable tests for software available in EESSI
 - Running on selected (single node) tests when building new software for EESSI (before deployment)
 - Periodically (daily/weekly) on about multiple different systems
 - Can also be used for other software stacks (that are built with EasyBuild)
 - Periodic runs of EESSI test suite help to catch performance regressions

-
- EESI / LAMMPS_Ij (study "default", 25/08/2026 - 11/09/2026)**
Software stack=production
- The chart displays the evolution of the software stack for the EESI / LAMMPS_Ij study in the production environment. The Y-axis represents a numerical value (likely a version or identifier) ranging from 0 to 9,000. The X-axis represents time, with labels for 27/08/26, 29/08/26, 31/08/26, 01/09/26, 03/09/26, 05/09/26, 07/09/26, 09/09/26, and 11/09/26. A vertical dashed line marks the date 31/08/26, indicating a significant update event.
- System Legend:**
- accelgor:accelgor (Blue line)
 - doduo:doduo (Orange line)
 - donphan:donphan (Light Green line)
 - gallade:gallade (Dark Green line)
 - jotik:jotik (Pink line)
 - litleo:litleo (Red line)
 - Magic_Castle:aarch64 (Light Orange line)
 - Magic_Castle:aarch64 (Orange line)
 - Magic_Castle:aarch64 (Purple line)
 - Magic_Castle:x86_64 (Dark Purple line)
 - Magic_Castle:x86_64 (Light Blue line)
 - Magic_Castle:x86_64 (Blue line)
 - shinx:shinx (Light Green line)
 - skiddo:skiddo (Dark Green line)
- Environment Legend:**
- default (Open circle)
- Key observations from the chart:
- The **litleo:litleo** system (Red line) shows a major update on 31/08/26, jumping from approximately 3,500 to 8,000.
 - The **accelgor:accelgor** system (Blue line) also shows a significant update on 31/08/26, jumping from approximately 3,500 to 6,300.
 - The **donphan:donphan** system (Light Green line) shows a major update on 31/08/26, jumping from approximately 3,500 to 8,000.
 - The **gallade:gallade** system (Dark Green line) shows a major update on 31/08/26, jumping from approximately 3,500 to 8,000.
 - The **shinx:shinx** system (Light Green line) shows a major update on 31/08/26, jumping from approximately 3,500 to 8,000.
 - The **skiddo:skiddo** system (Dark Green line) shows a major update on 31/08/26, jumping from approximately 3,500 to 8,000.
 - The **Magic_Castle** systems (various lines) show minor updates and fluctuations throughout the period.

Building on top of EESSI improvements



With EasyBuild

- Just module load EESSI-extend!

Wrapper module to set up all the environment for you

Without Automated build tools

- Load `buildenv` module
- Procedure is documented
- Most cases untested and undocumented, we appreciate feedback!

Manually building software on top of EESSI (without EasyBuild)

Warning

Even with the help of the `buildenv` module, this documentation is intended for "experts" in the runtime linker and its behaviour, and most cases are untested. Any feedback on this topic is highly appreciated.

Building and running software on top of EESSI without EasyBuild is now more straightforward thanks to the freshly developed `buildenv` module.

EESSI uses a [compatibility layer](#) to ensure that it takes as few libraries from the host as possible. Starting an EESSI prefix shell before building and loading any of the `buildenv` modules (one per `foos` toolchain), this is the safest way to make sure all libraries will point to the required locations in the compatibility layer and do not leak in from the host operating system. A step by step:

With Spack...? 🤔

<https://www.eessi.io/docs/blog/2026/02/05/Spack-on-top-of-EESSI-best-of-both-worlds>

https://www.eessi.io/docs/using_eessi/building_on_eessi



software.eessi.io (production)

- Permanent installations
- Only install releases
- Builds for all CPU architectures
- `/cvmfs/software.eessi.io/version/`

```
/cvmfs/dev.eessi.io/versions/2023.06/software
```

```
-- linux
  -- x86_64
    -- amd
      -- zen2
        -- modules
          -- all
            -- ESPResSo
              -- 4.2.2-foss-2023a-2ba17de6096933275abec0550981d9122e4e5f28.lua
            -- GROMACS
              -- 2024.3-foss-2023b-d0f934abfd1394621c40858a2c2dd9123451df4e.lua
            -- LAMMPS
              -- 570c9d190fee556c62e5bd0a9c6797c4dffcc271-foss-2023a-kokkos-dev_OBMD.lua
          -- ...
```

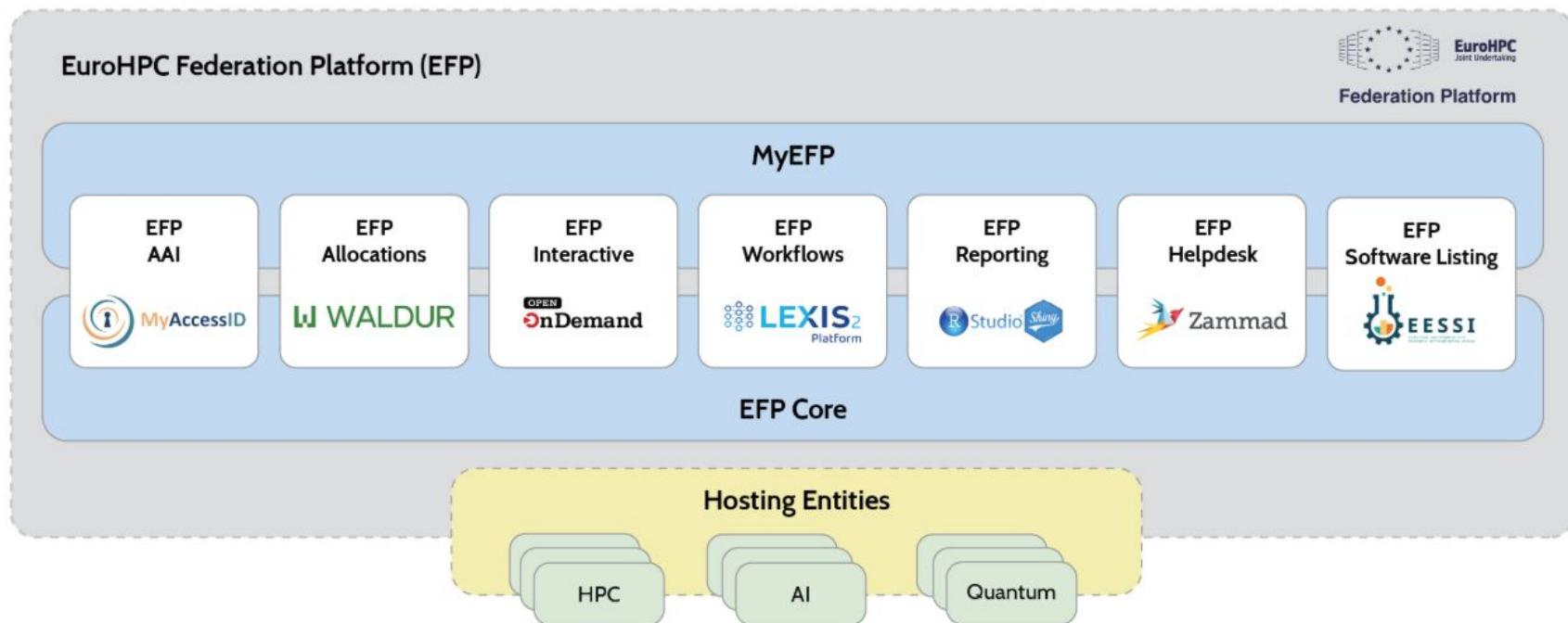
dev.eessi.io

- Temporary installations
- Can install from commits
- Patches can be added
- Can skip some CPU targets
- dev.eessi.io built on top of software.eessi.io
- Subdirectories per project
- `/cvmfs/dev.eessi.io/project/version/`

Currently only available for codes
within the MultiXscale project!

Further integrations

European Federation Platform (EFP)



<https://my-eurohpc.eu>

EFP webinar series in Oct'26: <https://docs.my-eurohpc.eu/training/overview>

European Open Science Cloud (EOSC)



October 22, 2025 · 5 min read

EESSI available on EOSC EU Node



EESSI has recently been working together with the [European Open Science Cloud \(EOSC\)](#) to provide easy access to EESSI for those using EOSC compute resources.

A first implementation of EESSI support is now available as the [EESSI Compute Node tool in the EOSC EU Node Tools Hub](#).

[Continue reading](#)

Tools Hub

The place to find, create, publish and run tools.

[About Tools Hub](#)

+ Create

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Q Search

All Tools

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Favourites

Deployments

Alan O'Cais

EESSI Compute Node

Deploy a compute node with an extra HD disk, EESSI (<https://eessi.github.io/docs>) installed, getting the IP and SSH credentials to access via ssh.

Status: Published

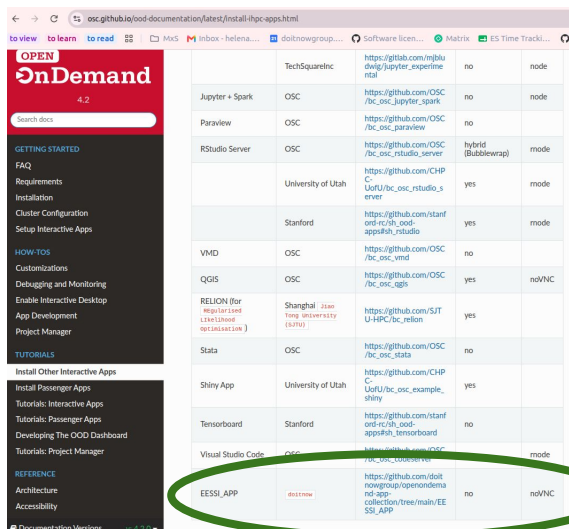


Deploy

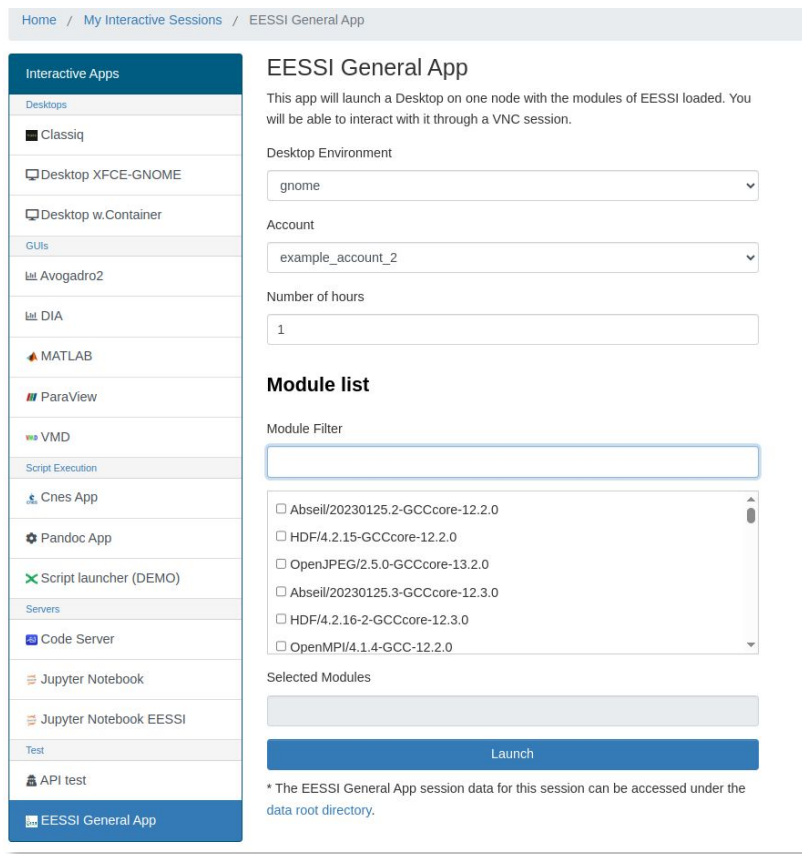
<https://eessi.io/docs/blog/2025/10/22/eosc/>

Open OnDemand (OOD)

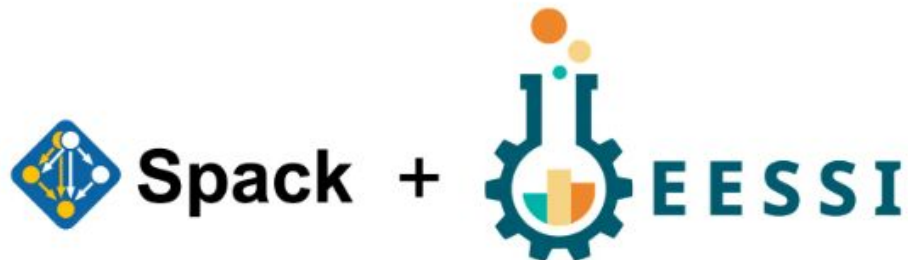
- Web Desktop Based application
- Modules field to select software needed by the user
- Selected modules unwritable field to check the selected ones
- Search mechanism to find specific modules



<https://osc.github.io/ood-documentation/latest/install-ihpc-apps.html>



Using Spack on top of EESSI: A Proof of Concept



EESSI provides a rich set of software installations, which were built and installed using [EasyBuild](#).

Can we convince [Spack](#) to reuse software installed in EESSI, and by doing so make EESSI more attractive to those of you who prefer using Spack over EasyBuild?

Several people have expressed interest in exposing the software installations provided by EESSI to Spack, so they can be used as dependencies when building and installing software with Spack.

<https://eessi.io/docs/blog/2026/02/05/Spack-on-top-of-EESSI-best-of-both-worlds>

Workflow integrations



- Fresh out of the oven!
- Snakemake plugin
 - <https://github.com/EESSI/snakemake-plugin>
 - Ready for testing...
- Nextflow plugin: work in progress...

Trainings & dissemination

EESSI Happy Hours

<https://eessi.io/docs/training-events/happy-hours-sessions>



EESSI Happy Hour sessions

Come join us
every ~~Monday~~
month!!!!!!



provided by [EuroHPC CoE MultiXscale](#)

What if you no longer have to install a broad range of scientific software from scratch on every laptop, HPC cluster, or cloud instance you use or maintain, without compromising on performance?

The European Environment for Scientific Software Installations [EESSI](#) comes to the rescue!

Join us for EESSI Happy Hour, an informal, weekly online session to explore and discuss key EESSI topics and updates with the community.

Each series will focus on a specific main topic over several sessions, giving participants the opportunity to ask questions, get hands-on guidance, and share experiences.



Where? Online, via Zoom ([click here to join!](#))



When? **Mondays, 14:00-15:00 CE(S)T**



Who should join? Everyone interested in EESSI.

Older sessions
available in YouTube
[https://www.youtube.com/
@eessi_community](https://www.youtube.com/@eessi_community)

Webinars series - spring 2026

- Introduction to EESSI
- Building software on top of EESSI
+ contributing to EESSI
- Using EESSI for Continuous Integration
- Introduction to CernVM-FS
- Using EESSI as a base for a central software stack



The screenshot shows a workflow diagram for adding software to EESSI. It starts with a 'Contributor' icon, followed by 'Create PR', 'EESSI software team', 'Review PR', 'Merge PR', 'Build & test', 'Deploy to production', and 'Release'. A central box labeled 'How to add software to EESSI' is connected to the 'Create PR' and 'Review PR' steps. The diagram is titled 'Semi-automated workflow for adding software to EESSI' and includes a URL 'eessi.io/docs/software_layer/adding_software' at the bottom.

EESSI webinar series - spring 2026

by EESSI

Playlist • 5 videos • 45 views

<https://www.eessi.io/docs/training-events/2026/webinar-series-2026Q2>

▶ Play all

<https://eessi.io/docs/training-events/2026/webinar-series-2026Q2/>

Hackathons, and other events



Copenhagen, EuroHPC User days 2025

Spotify podcast!↓



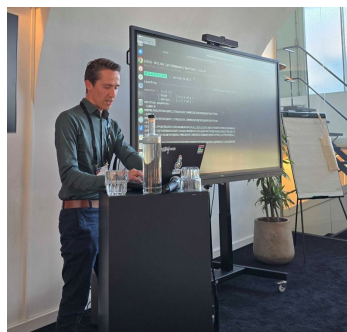
HPCKP 2026, Bcn



ISC 2026, Hamburg



RSE Days Belgium 2025



SURF Research days 2025



Hackathon + tutorial 2026, CINECA

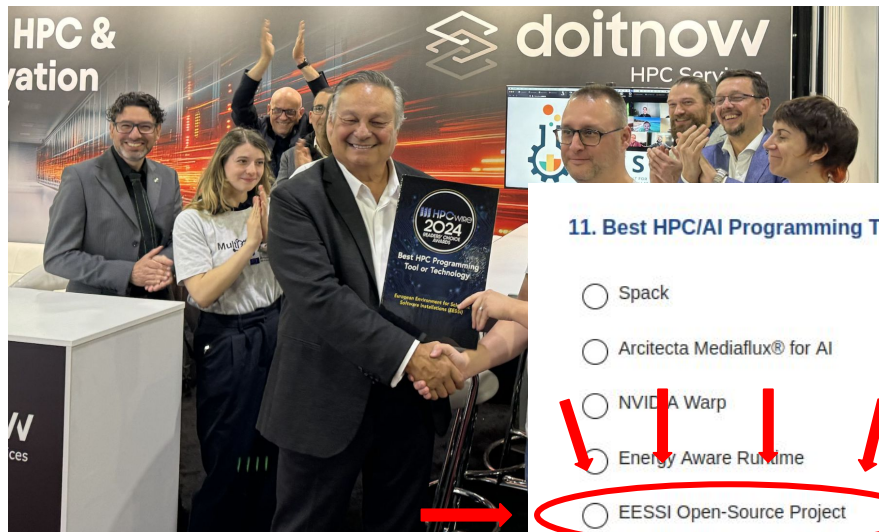


EUM 2026, Guimarães

And much more to come!



EESSI won (and is nominated again to) for an HPCWire Readers' Choice award!



11. Best HPC/AI Programming Tool or Technology

- ☐ Spack
- ☐ Arcitector Mediaflux® for AI
- ☐ NVIDIA Warp
- ☐ Energy Aware Runtime
- ☒ EESSI Open-Source Project
- ☐ AWS Agentix IDE
- ☐ NVIDIA cuBLAS
- ☐ MET Norway and AI improving long-range weather prediction

Help us win!



<https://eessi.io/docs/blog/2026/09/20/hpcwire-readers-choice-awards-2026>

Support for installing, using, contributing to EESSI



eessi.io/docs/support

- **Via GitLab, or via email: support@eessi.io**
- Report problems
- Ask questions
- Request additional software
- Get help with contributing to EESSI
- Suggest enhancements, additional features, ...
- Confidential tickets possible (security issues, ...)

Search or go to...

Project

- EESSI support portal
- Manage
- Plan
- Code
- Build
- Deploy
- Operate
- Monitor
- Analyze

Help

EESSI / EESSI support portal

README.md

EESSI support portal

MultiXscale **E E S S I**
EUROPEAN ENVIRONMENT FOR SCIENTIFIC SOFTWARE INSTALLATIONS

Thanks to the [MultiXscale EuroHPC project](#) we are able to provide support to the u

Contact

Create an issue with your GitLab account

If you have a GitLab account or create one you can create and manage your issue also use one of our issue templates.

Contact us via E-mail

If you do not have a GitLab account you can also ask for support via E-mail.

Dedicated support team, thanks to EuroHPC Centre-of-Excellence

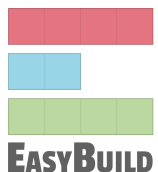




Installing
software
on top of

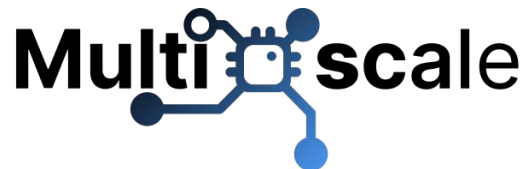
EESI

EUROPEAN ENVIRONMENT FOR
SCIENTIFIC SOFTWARE INSTALLATIONS

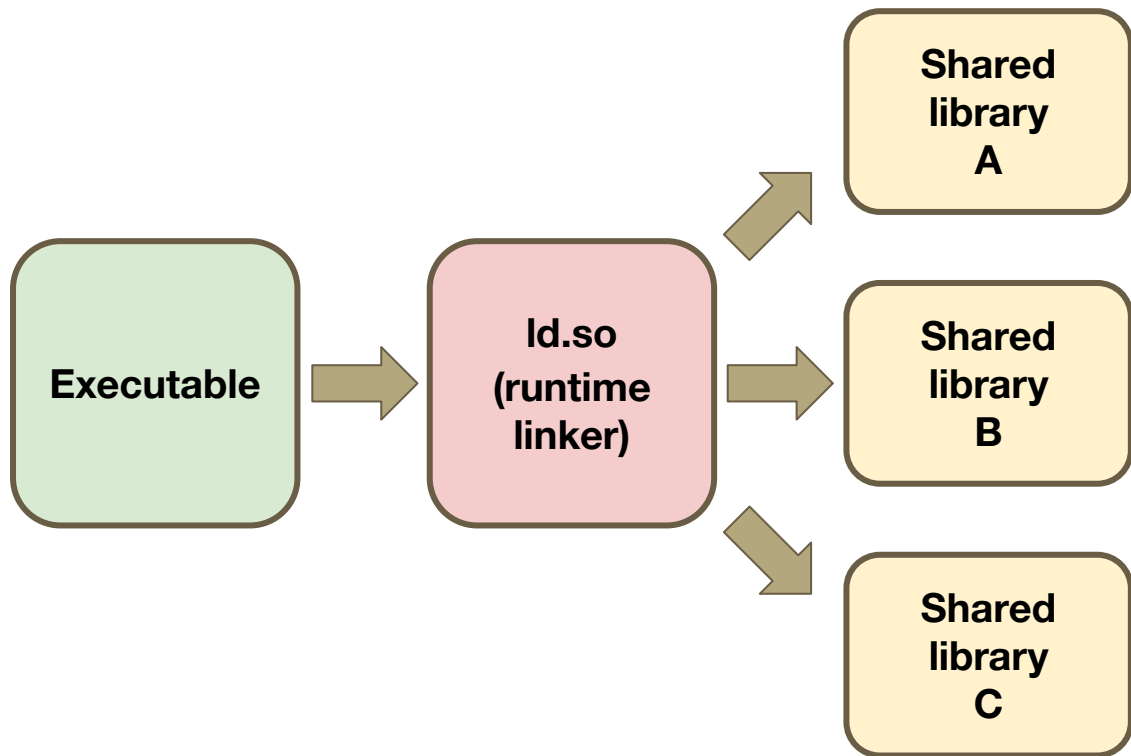


Fri 25 September 2026

Alan O'Cais (freelancer for Ghent University)



How does a program get loaded onto a computer?



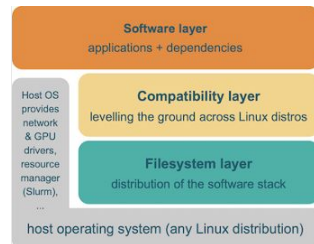
Where does the loader find the libraries?

- Clues in environment variables
- Information in the binary (RPATH)
- It's own defaults

The “trick” behind the EESSI software layer

Software installations only link to **libraries from compatibility layer or other installations in software layer**, *not* from host (Linux) operating system.

- Because of this, these software installations work on any Linux distribution
- This is ensured via *RPATH linking*
- Build tools (like CMake) need to be made aware of alternative system root (sysroot)
- Some exceptions need to be made: GPU driver libraries must be provided by host OS
- “Containers without the containing”



Example: `gmx` binary for GROMACS in EESSI

- Let's inspect the `gmx` binary in a GROMACS installation up close...
- We'll use standard tools like `readelf` and `ldd` to show that:
 - RPATH section of binary specifies hardcoded list of paths for libraries
 - `$LD_LIBRARY_PATH` is irrelevant (actually, only used as a fallback)
 - All libraries used by `gmx` binary come from EESSI compatibility layer or dependencies provided via EESSI software layer
- Everything shown is standard on Linux, no tricks up our sleeve

Preparing environment to inspect `gmx` binary for GROMACS



- Initialize shell for using EESSI version 2025.06:

```
source /cvmfs/software.eessi.io/versions/2025.06/init/lmod/bash
```

- Find and load a GROMACS module:

```
module avail GROMACS
```

```
module load GROMACS/2026.2-foss-2025b
```

- Check path to `gmx` binary:

```
command -v gmx    (or which gmx)
```

Inspect RPATH section of `gmx` binary with `readelf`

`readelf -d` command reveals details for a specific binary:

```
readelf -d $(command -v gmx)
```

- (RPATH) section specifies which paths should be considered first when loader looks for libraries that are required by `gmx` binary
- (NEEDED) entries specify which libraries are required, which themselves may require additional libraries in turn

Inspect RPATH section of `gmx` binary with `readelf`

There are different types of paths in (RPATH) section:

- Paths to **subdirectories of `host_injections`** (`/rpath_overrides/`)
which allow to inject specific libraries (via a path controlled by system administrators)
- Paths for **libraries in software installation directory** itself (like `/lib`)
- Paths **relative to path of binary** itself (`$ORIGIN/...`)
- Paths to libraries provided by **dependencies in EESSI software layer** (`/software/`)
- Paths to libraries provided by **EESSI compatibility layer** (`/compat/`)

Inspect libraries used by `gmx` binary with `ldd`

`ldd` command reveals libraries that are used when launching `gmx` binary:

```
ldd $(command -v gmx)
```

All libraries are provided either by EESSI compatibility layer or software layer.

Some noteworthy ones:

GROMACS' own library (`libgromacs.so`), provided by the GROMACS software installation directory:

```
libgromacs.so.10 => /cvmfs/software.eessi.io/.../software/GROMACS/.../lib/libgromacs.so.10 (0x000014f646e00000)
```

MPI library (`libmpi.so`), provided by OpenMPI dependency in EESSI software layer:

```
libmpi.so.40 => /cvmfs/software.eessi.io/.../software/OpenMPI/.../lib64/libmpi.so.40 (0x000014f646a00000)
```

GNU C library (`glibc`, `libc.so`), provided by EESSI compat layer (*not* by host OS):

```
libc.so.6 => /cvmfs/software.eessi.io/.../compat/.../lib/.../lib64/libc.so.6 (0x000014f646425000)
```

Building software on top of EESSI



To build software on top of EESSI, we have to make sure that:

- No libraries from host OS are relied on, so that binaries are “portable” across Linux systems
- Path to required libraries is “hardcoded” into binaries and libraries, using RPATH linking.

This is a technical challenge, and is different from how people usually build software:

- At build time, specify library paths via `-L` option in compiler command or via `$LIBRARY_PATH`
- At run time, rely on standard system paths (`/usr/lib`) or `$LD_LIBRARY_PATH`

This can't be relied on when building software to be used in or on top of EESSI (too brittle).

Building software on top of EESSI: options

There are different options to build software on top of EESSI:

- Using **EasyBuild**
 - Should be configured correctly for building on top of EESSI
- Using a “**manual**” build procedure
 - Need to make sure that compatibility layer is used and that RPATH linking is done
- Using **Spack**
 - Can use use software provided by EESSI as “external packages”,
for compiler, dependencies

Building software on top of EESSI: options

There are different options to build software on top of EESSI:

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Building software on top of EESSI with EasyBuild

Over the last couple of years, significant enhancements have been made in EasyBuild to make it easier to build software such that it can be included in EESSI, including:

- Improve support for RPATH linking, and make it the default in EasyBuild v5.0
- `--sysroot` configuration option, and making various easyblocks aware of it
- Support for [easystack files](#)
- `--sanity-check-only` option
- `--from-commit` option, which is more reproducible than `--from-pr`
- CUDA sanity check

For a recent introduction to EasyBuild, see [Introduction to EasyBuild webinar \(spring 2025\)](#)



Building software on top of EESSI with `EESSI-extend`



- An `EESSI-extend` module is provided by EESSI which loads + pre-configures EasyBuild
- Very useful when doing local installations on top of EESSI with EasyBuild, or testing
- Basic usage:
 - Initialize EESSI
 - (optional) Set `$EESSI_*_INSTALL` environment variable to control installation path
 - Load `EESSI-extend` module
 - Check EasyBuild configuration (`eb --show-config`)
 - Provide one or more `easyconfig` files to EasyBuild to install software on top of EESSI
- The EESSI build-deploy bots also use `EESSI-extend`, so this mechanism is very well tested!
- Detailed documentation available at https://eessi.io/docs/using_eessi/building_on_eessi

Building software on top of EESSI with EESSI-extend



Hands-on demo first and then try for yourself!

EESSI 2025.06: re-installing imageio with foss/2024a

imageio-2.36.1-gfbf-2024a.eb is already available in EESSI/2025.06 , let's rebuild it:

```
source /cvmfs/software.eessi.io/versions/2025.06/init/lmod/bash
```

```
module load EESSI-extend
```

```
eb --show-config
```

```
eb --rebuild imageio-2.36.1-gfbf-2024a.eb
```



Building software on top of EESSI with EESSI-extend

Configuring EESSI-extend before loading via `$EESSI_*_INSTALL` env. variables

- `$EESSI_USER_INSTALL` to specify that **user-specific** installation prefix that should be used
 - Permissions are set such that directories are only readable by user (umask 077)
- `$EESSI_PROJECT_INSTALL` to specify that **project** installation prefix that should be used
 - Permissions set accordingly: GID bit, group writable installation directories, umask 002, ...
- `$EESSI_SITE_INSTALL` can be set to help with **site-wide** installations on top of EESSI
- `$EESSI_CVMFS_INSTALL` can be set to specify that installation should be done in **CernVM-FS repo**

See also https://eessi.io/docs/using_eessi/building_on_eessi/#using-the-eessi-extend-module

Building software on top of EESSI with EESSI-extend



Hands-on demo (EESSI 2025.06): installing IQ-TREE 3.1.2

- No easyconfig file for IQ-TREE 3.1.2 available yet in EasyBuild 5.3.0, but a pull request was merged for it recently...
- Merged pull request to EasyBuild: <https://github.com/easybuilders/easybuild-easyconfigs/pull/25981>
- Can test using EESSI-extend and `eb --from-pr:`

```
source /cvmfs/software.eessi.io/versions/2025.06/init/lmod/bash
```

```
export EESSI_USER_INSTALL=$HOME/EESSI-2025.06
```

```
module load EESSI-extend
```

```
eb --from-pr 25981 --missing
```

```
eb --from-pr 25981 --robot
```

Building software on top of EESSI manually, using `buildenv`



Using EasyBuild to build/install software on top is not required, there are other options.

However, you should make sure that:

- RPATH linking is used, or you need to set `$LD_LIBRARY_PATH`
- Compilers and build tools are aware of the alternate sysroot (compatibility layer)

To facilitate this, EESSI provides a `buildenv` module for each (foss) toolchain which:

- Sets up the build environment similar to what EasyBuild uses (`$CC`, `$CFLAGS`, etc.)
- Injects RPATH wrappers for the compiler and linker commands (`gcc`, `g++`, `gfortran`, `ld`, ...)

Detailed documentation available at https://eessi.io/docs/using_eessi/building_on_eessi

Building software on top of EESSI manually



Hands-on demo: Manual build, with the help of `buildenv`

```
source /cvmfs/software.eessi.io/versions/2025.06/init/lmod/bash
```

```
module load buildenv/default-foss-2025b
```

```
module load libpng/1.6.50-GCCcore-14.3.0
```

```
git clone https://github.com/R-Gerard/libpng-demo
```

```
cd libpng-demo
```

```
make
```

```
readelf -d target/pngdemo
```

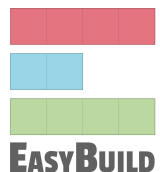
```
ldd target/pngdemo
```




Workflow for
adding additional
software to

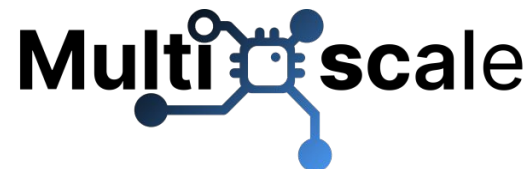
E E S S I

EUROPEAN ENVIRONMENT FOR
SCIENTIFIC SOFTWARE INSTALLATIONS



Fri 25 September 2026

Thomas Röblitz (University of Bergen)



Q&A via EESSI Slack

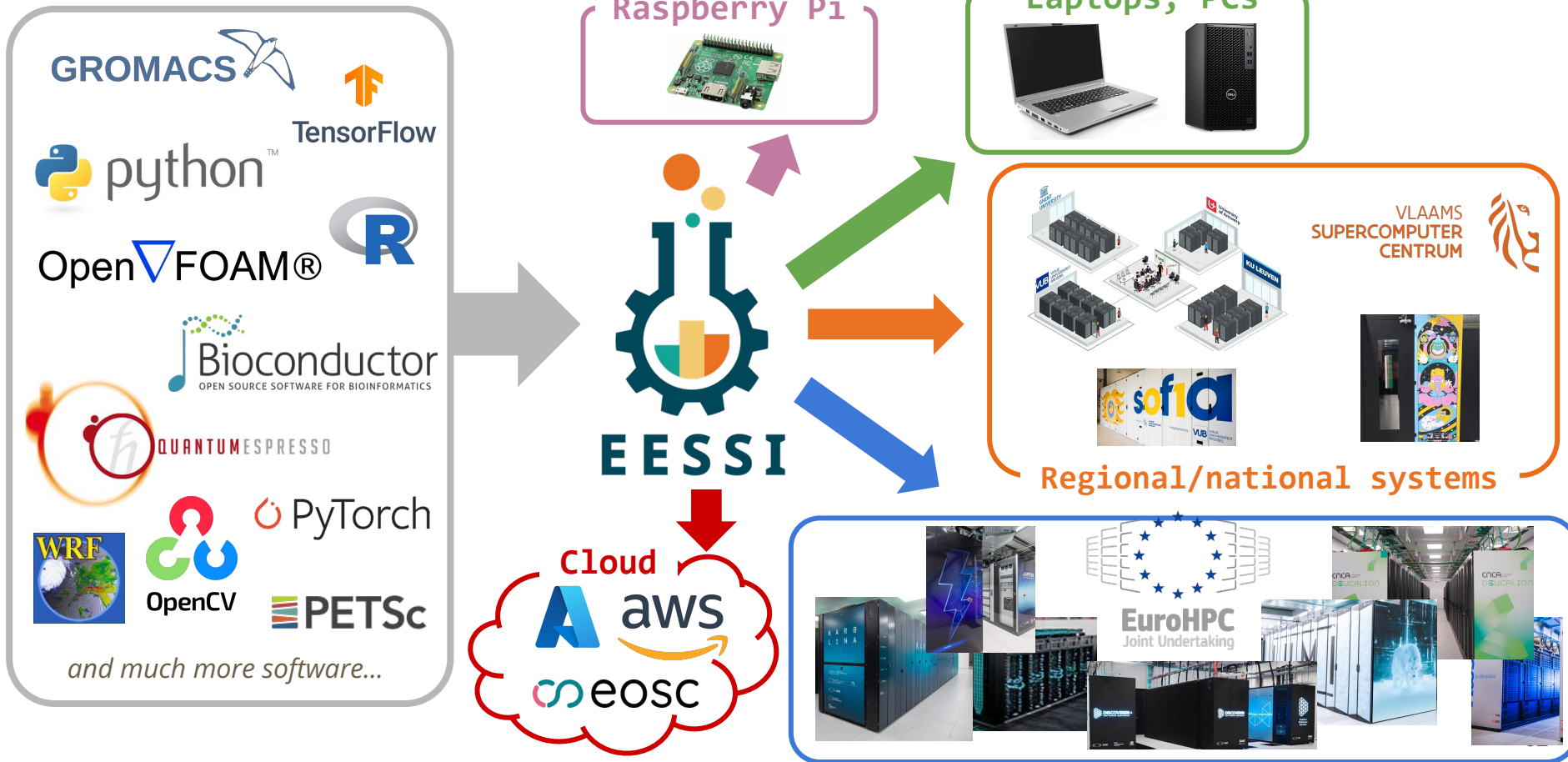


Please join the #eurohpc-user-days-2026-hackathon channel in the EESSI Slack for questions and discussion

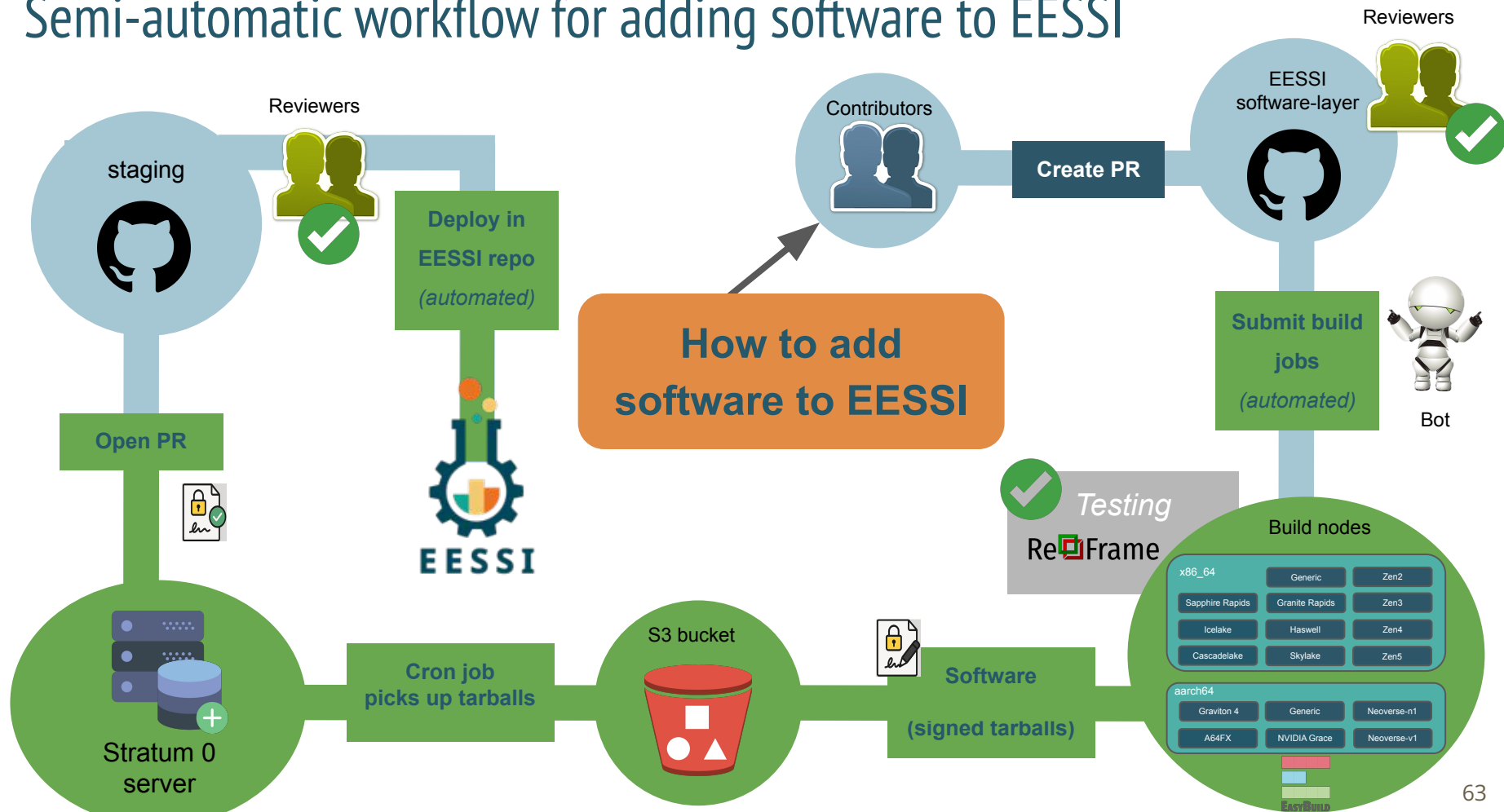
Step 1) **Join the EESSI Slack**, see “Slack channel” link at <https://eessi.io>

Step 2) **Join [#eurohpc-user-days-2026-hackathon](#) channel** in EESSI Slack
(direct link: <https://eessi-hpc.slack.com/archives/C0C3TJ0RCG7>)

Recap: EESSI as shared software stack for HPC, cloud, and beyond



Semi-automatic workflow for adding software to EESSI



Supported CPU targets

https://eessi.io/docs/software_layer/cpu_targets



- AMD, Intel (`x86_64`)
 - 6 generations of Intel CPUs:
Haswell, Skylake, Cascade Lake, Ice Lake, Sapphire Rapids, Granite Rapids
 - 4 generations of AMD CPUs:
Zen2 (Rome), Zen 3 (Milan), Zen4 (Genoa), Zen5 (Turin)
- Arm (`aarch64`)
 - 5 different microarchitectures:
Fujitsu A64FX, NVIDIA Grace, Neoverse N1 + V1, Graviton 4 (*new*)
- Plus a generic fallback for both `x86_64` and `aarch64`
- RISC-V CPUs are experimental (only `riscv64/generic`, only EESSI/2026.06)

Supported GPU targets (NVIDIA)

https://eessi.io/docs/software_layer/gpu_targets



- Different generations of NVIDIA GPUs, based on CUDA Compute Capability (CC), see also <https://developer.nvidia.com/cuda/gpus>
- In EESSI 2023.06: CC 7.0 (V100, T4), 8.0 (A100 & co, incl. L40), 9.0 (H100/H200)
- In EESSI 2025.06: CC 7.0, 8.0, 9.0, 10.0 (B100/B200/B300), 12.0 (B10, RTX PRO Blackwell)
- In EESSI 2026.06: **CC 7.5** (CUDA 13.x) , 8.0, 9.0, 10.0, 12.0
- GPU driver libraries (include `libcuda.so`) must be exposed to EESSI
- See https://www.eessi.io/docs/site_specific_config/gpu
- About 10% of all 40k software installations in EESSI are for (NVIDIA) GPUs

See also *NVIDIA + AMD GPU support in EESSI* talk @ EUM'26 ([slides](#), [recording](#))

Supported GPU targets (AMD/ROCm) https://eessi.io/docs/software_layer/gpu_targets



- First set of modules available in EESSI/2025.06

```
----- /cvmfs/software.eessi.io/versions/2025.06/software/linux/x86_64/generic/accel/amd/gfx908/modules/all -----  
Boost/1.88.0-rocm-compilers-19.0.0-ROCm-6.4.1 (g,D) rocBLAS/4.4.0-rocm-compilers-19.0.0-ROCm-6.4.1 (g)  
HIP/6.4.1-rocm-compilers-19.0.0-ROCm-6.4.1 (g) roc-cmake/0.14.0-rocm-compilers-19.0.0-ROCm-6.4.1 (g)  
hipBLAS-common/1.2.0-rocm-compilers-19.0.0-ROCm-6.4.1 (g) rocm-compilers/19.0.0-ROCm-6.4.1 .  
hipBLASLt/0.12.1-rocm-compilers-19.0.0-ROCm-6.4.1 (g) ROCm-LLVM/19.0.0-GCCcore-14.2.0-ROCm-6.4.1 .  
HIPIFY/19.0.0-rocm-compilers-19.0.0-ROCm-6.4.1 (g) rocminfo/1.0.0-rocm-compilers-19.0.0-ROCm-6.4.1 (g)  
msgpack-cxx/7.0.0-rocm-compilers-19.0.0-ROCm-6.4.1 (g) rocTracer/4.1.0-rocm-compilers-19.0.0-ROCm-6.4.1 (g)  
RCCL/2.22.3-rocm-compilers-19.0.0-ROCm-6.4.1 (g) Tensile/4.43.0-rocm-compilers-19.0.0-ROCm-6.4.1 (g)
```

- Support for different AMD GPU generations: [see docs for details](#)

gfx908: MI100, gfx90a: MI200, gfx942: MI300, gfx1030: RX6000, gfx1100/1101:

RX7000, gfx1200/1201: RX8000 and RX9000

- Initial goal: GROMACS with ROCm 6.4.1 (WIP), ROCm 7.x next

See also *NVIDIA + AMD GPU support in EESSI* talk @ EUM'26 ([slides](#), [recording](#))

Contributing software to EESSI



To contribute additional software to EESSI, the following must be taken into account:

- An easyconfig file must be available in EasyBuild to install the software
 - Either in the latest EasyBuild release, or in a (merged) pull request to EasyBuild
- A pull request must be opened to the `EESSI/software-layer` repository
- The EESSI contribution policy must be followed
 - See https://eessi.io/docs/adding_software/contribution_policy
 - Must be redistributable software (like open source), must work on all CPU targets, must be built by EESSI bot, should be recent software version and use recent toolchain, ...



Contributing software to EESSI, step-by-step

1. Find or create a (working) **easyconfig file** to install the software
2. **Test it in the EESSI build environment** (using `EESSI-extend`)
3. **Contribute it to EasyBuild** (if it hasn't been yet)
4. Open a **pull request** to the `EESSI/software-layer` repository
5. See if installation works on **all supported CPU targets**
 - Somebody from the EESSI “builders” team will trigger the bot to do test builds
6. When that all checks out, the software installations will be **deployed to EESSI**

Recommendations for contributing software to EESSI



- **Test locally first**, using `EESSI-extend`
 - Can be on multiple CPU targets or families, but not strictly required
- Always **use latest EasyBuild release** (see `module avail EasyBuild` in EESSI)
- Pull request to EESSI can be opened while `easyconfig` PR to EasyBuild is still open, but *deployment* to EESSI can only be done when EasyBuild PR has been *merged*
- Work with EESSI “builders” team to resolve any problems that may pop up
 - Via comments in pull requests
 - Via EESSI Slack (see `#software-layer` or `#support` channels), see link @ eessi.io

Recommendations for contributing software to EESSI



Example output for testing building a software locally

```
== COMPLETED: Installation ended successfully (took 5 mins 52 secs)
== Results of the build can be found in the log file(s)
/home/ec2-user/eessi/versions/2025.06/software/linux/aarch64/neoverse_n1/software/scikit-image/0.25.0-foss-2024a/easyb
uild/easybuild-scikit-image-0.25.0-20260519.160252.log.bz2
== Running post-easyblock hook...

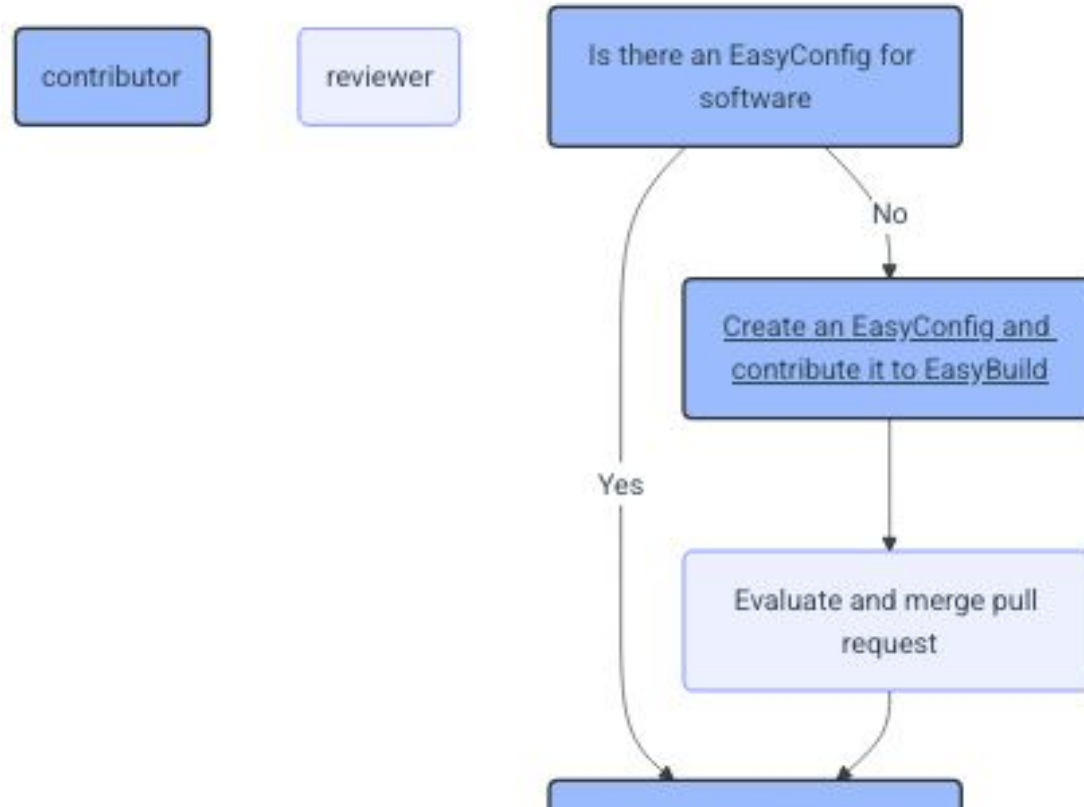
== Build succeeded for 5 out of 5 (total: 10 mins 13 secs)
== Summary:
* [SUCCESS] imageio/2.36.1-gfbf-2024a
* [SUCCESS] bokeh/3.6.0-gfbf-2024a
* [SUCCESS] dask/2024.9.1-gfbf-2024a
* [SUCCESS] PyWavelets/1.8.0-foss-2024a
* [SUCCESS] scikit-image/0.25.0-foss-2024a
== Temporary log file(s) /tmp/eb-13zsn0gu/easybuild-m2i6ku9x.log* have been removed.
== Temporary directory /tmp/eb-13zsn0gu has been removed.
{EESSI/2025.06} [ec2-user@ip-172-31-32-214 ~]$
```

All versions of scikit-image are already in EESSI/2025.06. Try with Cartopy-0.24.1-foss-2024a.eb

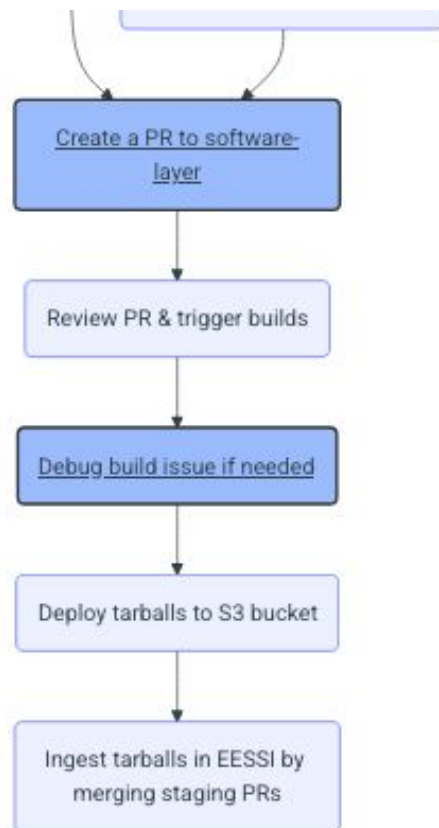
Detailed guidelines for contributing software to EESSI

- See https://eessi.io/docs/adding_software/overview
- Documentation for contributing to EasyBuild:
 - https://docs.easybuild.io/contributing/#contributing_easyconfigs
 - https://docs.easybuild.io/integration-with-github/#github_new_update_pr
 - <https://tutorial.easybuild.io/2023-eb-eessi-uk-workshop/easybuild-contributing>
- Step-by-step instructions for opening PR to EESSI:
https://www.eessi.io/docs/adding_software/opening_pr

Detailed guidelines for contributing software to EESSI



Detailed guidelines for contributing software to EESSI



Contributing software to EESSI [hands-on]



Hands-on (EESSI/2025.06): contributing `Cartopy/0.24.1` with `foss/2024a`

Local test installation

Contribution to EESSI via GitHub repository `EESSI/software-layer`

Contributing software to EESSI [hands-on]



Hands-on (EESSI/2025.06): contributing `Cartopy/0.24.1` with `foss/2024a`

Local test installation

1. Setup environment
2. Find EasyConfig
3. Determine status
4. Test installation locally

Contributing software to EESSI [hands-on]



Hands-on (EESSI/2025.06): contributing Cartopy/0.24.1 with foss/2024a

1. Setup environment

```
source /cvmfs/software.eessi.io/versions/2025.06/init/lmod/bash  
module load EESSI-extend/2025.06-easybuild
```


Contributing software to EESSI [hands-on]



Hands-on (EESSI/2025.06): contributing `Cartopy/0.24.1` with `foss/2024a`

2. Find EasyConfig

```
eb -S Cartopy
```

...

```
* $CFG$1/Cartopy-0.24.1-foss-2024a.eb
```

Contributing software to EESSI [hands-on]



Hands-on (EESSI/2025.06): contributing Cartopy/0.24.1 with foss/2024a

3. Determine status (two methods: `module avail` and `eb --missing` or `eb -M`)

```
module avail Cartopy
```

```
Cartopy/0.24.1-foss-2025a (same version but different toolchain)
```

```
eb -M Cartopy-0.24.1-foss-2024a.eb
```

```
...
```

```
4 out of 126 required modules missing:
```

- * pyproj/3.7.0-GCCcore-13.3.0 (pyproj-3.7.0-GCCcore-13.3.0.eb)
- * Shapely/2.0.6-gfbf-2024a (Shapely-2.0.6-gfbf-2024a.eb)
- * Fiona/1.10.1-foss-2024a (Fiona-1.10.1-foss-2024a.eb)
- * Cartopy/0.24.1-foss-2024a (Cartopy-0.24.1-foss-2024a.eb)

Contributing software to EESSI [hands-on]



Hands-on (EESSI/2025.06): contributing Cartopy/0.24.1 with foss/2024a

4. Test installation locally

```
eb -r Cartopy-0.24.1-foss-2024a.eb
```

```
... ..
```

```
== Build succeeded for 4 out of 4 (total: 6 mins 35 secs)
```

```
== Summary:
```

```
* [SUCCESS] pyproj/3.7.0-GCCcore-13.3.0
```

```
* [SUCCESS] Shapely/2.0.6-gfbf-2024a
```

```
* [SUCCESS] Fiona/1.10.1-foss-2024a
```

```
* [SUCCESS] Cartopy/0.24.1-foss-2024a
```

Contributing software to EESSI [hands-on]



Hands-on (EESSI/2025.06): contributing `Cartopy/0.24.1` with `foss/2024a`

If test installation of `Cartopy-0.24.1-foss-2024a.eb` is OK,
we can open PR to `EESSI/software-layer` repository:

5. Add entry to

```
easystacks/software.eessi.io/2025.06/eessi-2025.06-eb-5.4.0-2024a.yml
```

6. Open pull request to `EESSI/software-layer` repository

7. Check if installation works on all CPU targets + get it deployed...

Break until ~ 13:40 Ireland time / ~ 14:40 CEST

Contributing software to EESSI [hands-on]



Hands-on (EESSI/2025.06): contributing `Cartopy/0.24.1` with `foss/2024a`

5. Add entry to

```
easystacks/software.eessi.io/2025.06/eessi-2025.06-eb-5.4.0-2024a.yml
```

```
git clone git@github.com:EESSI/software-layer.git
```

```
cd software-layer
```

```
git remote add YOU git@github.com:YOU/software-layer.git
```

Contributing software to EESSI [hands-on]



Hands-on (EESSI/2025.06): contributing Cartopy/0.24.1 with foss/2024a

5. Add entry to `easystacks/software.eessi.io/2025.06/eessi-2025.06-eb-5.4.0-2024a.yml`

...

```
git checkout -b add_cartopy_0_24_1_foss_2024a origin/main
```

```
vi easystacks/software.eessi.io/2025.06/eessi-2025.06-eb-5.4.0-2024a.yml
```

add the line

```
- Cartopy-0.24.1-foss-2024a.eb
```

```
git commit -am "{2025.06}[foss/2024a] Cartopy 0.24.1"
```

```
git push YOU add_cartopy_0_24_1_foss_2024a
```

Contributing software to EESSI [hands-on]



Hands-on (EESSI/2025.06): contributing `Cartopy/0.24.1` with `foss/2024a`

6. Open pull request to `EESSI/software-layer` repository

Use the link shown in the output of the last command (`git push ...`)

7. Check if installation works on all CPU targets + get it deployed...

Contributing software to EESSI [hands-on]



Hands-on (EESSI/2026.06): contributing latest scikit-bio (0.7.4) with `foss/2026.1`

scikit-bio 0.7.4 was released on September 21, 2026: no easyconfig for this version exists yet

1. **Create easyconfig** `scikit-bio-0.7.4-foss-2026.1.eb`
2. Test installation locally
3. **Create PR for EasyBuild easyconfig**
4. Add entry to `easystacks/software.eessi.io/2026.06/eessi-2026.06-eb-5.4.0-2026.1.yml`
5. **Use `from-commit` option** to let EasyBuild use easyconfig file(s) from easyconfigs pull request
6. Open pull request to `EESSI/software-layer` repository
7. Check if installation works on all CPU targets
8. Make sure that easyconfig PR gets merged, and then deployment in EESSI can be done

Contributing software to EESSI [hands-on]



Hands-on (EESSI/2026.06): contributing latest scikit-bio (0.7.4) with `foss/2026.1`

1. Create easyconfig `scikit-bio-0.7.4-foss-2026.1.eb`

- create from scratch
- ask some fAInd and verify resulting easyconfig
- double-check there is no open PR for it yet
- start from an existing easyconfig

```
eb -S scikit-bio
```

...

```
* $CFGS1/scikit-bio-0.7.2-foss-2025b.eb
```

Contributing software to EESSI [hands-on]



Hands-on (EESSI/2026.06): contributing latest scikit-bio (0.7.4) with `foss/2026.1`

1. Create easyconfig `scikit-bio-0.7.4-foss-2026.1.eb`

- start from an existing easyconfig

```
eb --copy-ec scikit-bio-0.7.2-foss-2025b.eb
```

```
cp scikit-bio-0.7.2-foss-2025b.eb scikit-bio-0.7.4-foss-2026.1.eb
```

```
chmod u+w scikit-bio-0.7.4-foss-2026.1.eb
```

Contributing software to EESSI [hands-on]



Hands-on (EESSI/2026.06): contributing latest scikit-bio (0.7.4) with `foss/2026.1`

1. Create easyconfig `scikit-bio-0.7.4-foss-2026.1.eb`
 - **!!!** study release notes & CHANGELOG for changes (dependencies & extensions)
 - change toolchain: 2025b -> 2026.1
 - change version: 0.7.2 -> 0.7.4
 - walk through dependencies: use `module avail` or `eb -S` to determine fitting version
update version for each (build) dependency accordingly
if no version for toolchain 2026.1 exists, you need to create a new easyconfig

Contributing software to EESSI [hands-on]



Hands-on (EESSI/2026.06): contributing latest scikit-bio (0.7.4) with foss/2026.1

1. Create easyconfig `scikit-bio-0.7.4-foss-2026.1.eb`

- walk through extensions (`exts_list`):

check for new versions on <https://pypi.org>

- inject checksums

```
eb scikit-bio-0.7.4-foss-2026.1.eb --inject-checksums --force
```

Contributing software to EESSI [hands-on]



Hands-on (EESSI/2026.06): contributing latest scikit-bio (0.7.4) with `foss/2026.1`

1. Create easyconfig `scikit-bio-0.7.4-foss-2026.1.eb`
 - follow all the steps for `biom-format` and `statsmodels` (no easyconfigs for those yet)
2. Test build these locally

```
eb -M biom-format-2.1.17-foss-2026.1.eb
```

```
eb -r biom-format-2.1.17-foss-2026.1.eb
```

```
eb -M statsmodels-0.15.0-gfbf-2026.1.eb
```

```
eb statsmodels-0.15.0-gfbf-2026.1.eb
```

Contributing software to EESSI [hands-on]



Hands-on (EESSI/2026.06): contributing latest scikit-bio (0.7.4) with `foss/2026.1`

3. Open Pull Request to GitHub repo `easybuilders/easybuild-easyconfigs`

- (set up and) use [EasyBuild's GitHub integration](#)
- `eb --preview-pr biom-format-2.1.17-foss-2026.1.eb`
- `eb --new-pr biom-format-2.1.17-foss-2026.1.eb`

Contributing software to EESSI [hands-on]



Hands-on (EESSI/2026.06): contributing latest scikit-bio (0.7.4) with `foss/2026.1`

scikit-bio 0.7.4 was released on September 21, 2026: no easyconfig for this version exists yet

4. Add entry to `easystacks/software.eessi.io/2026.06/eessi-2026.06-eb-5.4.0-2026.1.yml`

5. **Use `from-commit` option** to let EasyBuild use easyconfig file(s) from easyconfigs pull request

```
- biom-format-2.1.17-foss-2026.1.eb:
```

```
  options:
```

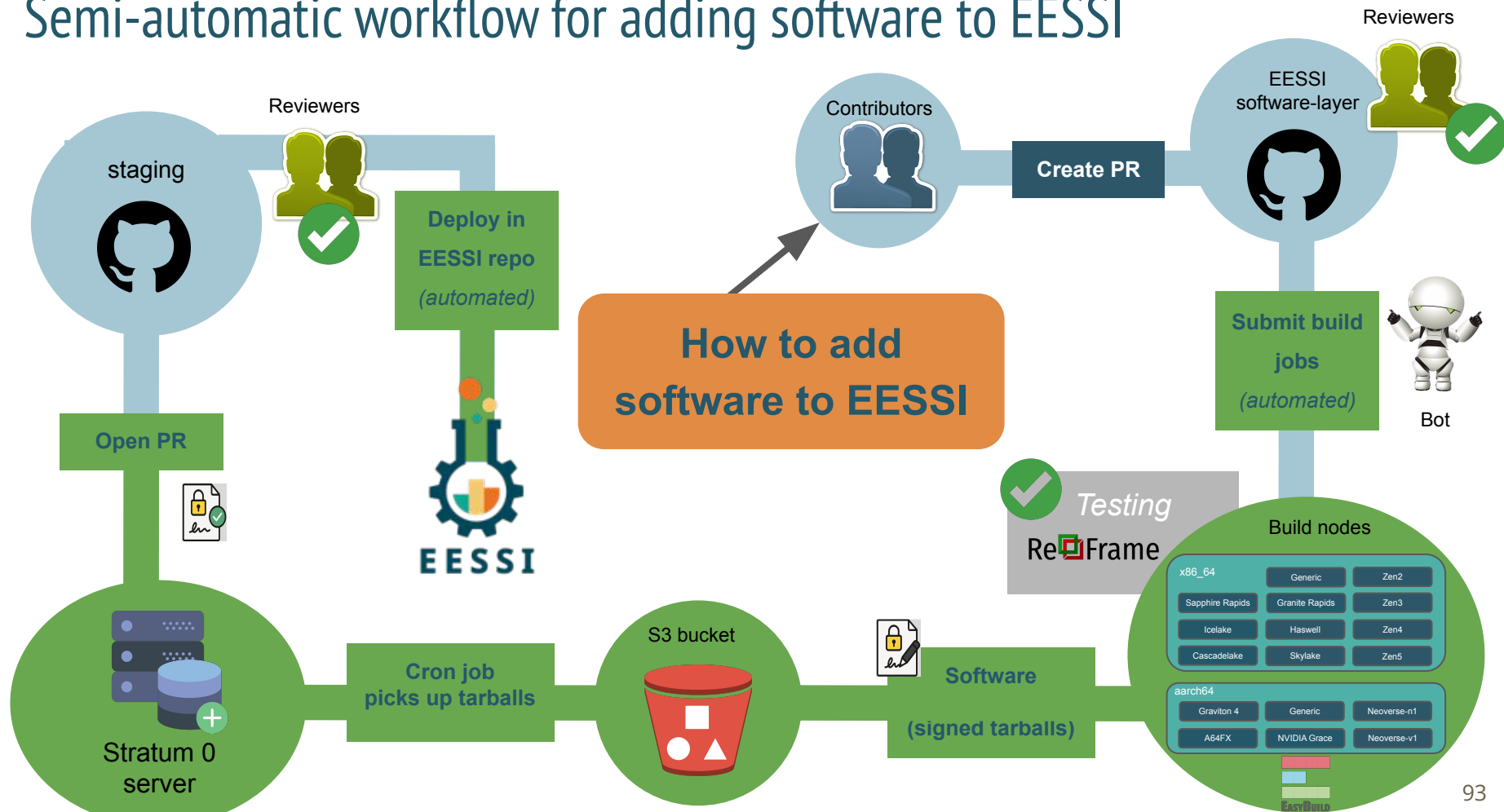
```
    from-commit: SHA_OF_COMMIT_ON_GITHUB
```

6. Open pull request to EESSI/software-layer repository

7. Check if installation works on all CPU targets

8. Make sure that easyconfig PR gets merged, and then deployment in EESSI can be done

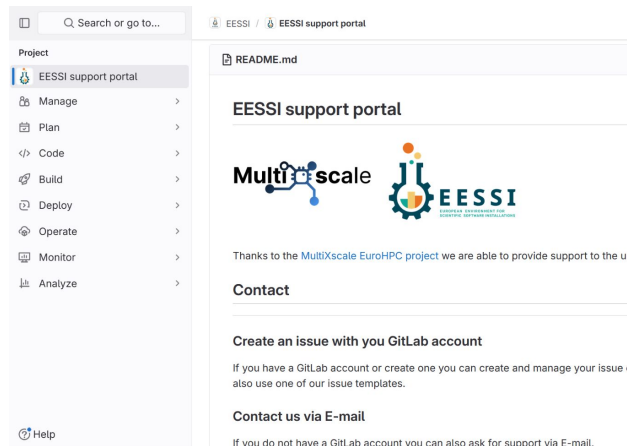
Semi-automatic workflow for adding software to EESSI



Support for installing, using, contributing to EESSI

- Via GitLab, or via email: support@eessi.io
- Report problems
- Ask questions
- Request additional software
- Get help with contributing to EESSI
- Suggest enhancements, additional features, ...
- Confidential tickets possible (security issues, ...)

eessi.io/docs/support



Dedicated support team, thanks to EuroHPC Centre-of-Excellence

